MILITARY SPECIFICATION

PELLET, ABSORBENT, COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of sterile cotton pellets for use in dental procedures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

JJJ-C-561	Cotton, Sterile (Purified) and Nonsterile
MMM-A-250	Adhesive, Water-resistant, (For Sealing Fiberboard Boxes)
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water Resistant (For Carton Sealing)

MILITARY

MIL-S-36586	Sterilization Test Strip Set, Bacterial Spore

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. PHARMACOPEIAL CONVENTION, INCORPORATED (USP)
Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

3. REQUIREMENTS

3.1 Material. The cotton pellets shall be made from purified cotton conforming to the requirements for grade "A" absorbent cotton of JJJ-C-561.

3.2 Construction and size. The cotton pellets shall be spun spherical in shape and shall have a nominal diameter of ¼ inch. The net weight of the pellets in the unit package (see 5.1.2.1) shall be a minimum of 500 grains (32.40 grams) when tested as specified in 4.4.1.

3.3 Sterility. The cotton pellets shall be sterilized by steam sterilization and shall be sterile when tested with a biological indicator as specified in 4.4.2.

3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (steam).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product.)
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.4 Workmanship. The pellets shall be free from dirt, lumps, and other impurities or foreign matter, and any other defects which may affect their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available, to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the cotton, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for examination purposes shall be one pellet or one unit package as indicated in table III. Not more than ten pellets shall be taken from each unit package.

Table I. Sampling for examination.

	Inspection level	AQL (percent defective)
For Visual Examination		
Major Defects	I	1.0
Minor Defects	S4	1.0
For Dimensional Examination	S3	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. The unit of product for testing purposes shall be one unit package.

Table II. Sampling for tests.

Characteristic	Requirement	Test Procedure	Inspection Level	AQL (percent defective)
Weight	3.2	4.4.1	S2	*
Sterility	3.3	4.4.2	S3	*

*Acceptance number shall be zero for all sample sizes.

4.3 Examination. The pellets shall be examined to determine compliance with all requirements of this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III. Examination shall not be restricted to the possible defects listed in the table.

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Table III. Classification of defects

Categories	Defects
A. Unit of Product	Unit package
Major	
101	Unit package not properly sealed to maintain sterility of contents
B. Unit of Product	Pellet
Minor	
201	Pellets not spun spherical in shape.
202	Pellets not free of dirt, lumps, and other impurities or foreign matter.

4.3.2 Dimensional examination. The cotton pellets shall be examined for defects in dimension. The unit of product shall be one pellet. Not more than five pellets shall be taken from each unit package.

4.3.3 Packaging inspection. The inspection of the package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptance number shall be zero.

Table IV. Examination of preparation for delivery

Examine	Defect
Contents	not as specified.
Container	not as specified.
Marking	Omitted; incorrect; illegible; improper location, sequence or method of application
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

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4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing, or banding shall be in accordance with the appendix of the box specification.

4.4 Tests.

4.4.1 Weight, net (unit package). The cotton pellets from the unit package shall be weighed on a balance weighing accurately to 0.001 gram, preferably an analytical balance. A balance calibrated to give the weight directly in grains or grams may be used.

4.4.2 Sterility test. The cotton pellets shall be tested for sterility in accordance with the method specified in the USP procedural details for sterility tests for purified cotton, gauze, surgical dressings, and related material using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.2.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II, Sampling Criteria for Tests.

4.4.2.1 Inoculated product. For steam sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.2.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be inoculated to be of equivalent resistance to spore populations included in MIL-S-36586. The method of incubation and number of test units shall be as indicated in 4.4.2 and 4.4.2.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.2.1.
NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

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4.4.2.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

5. PACKAGING.

5.1 Packaging requirements. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One box containing twenty five hundred cotton pellets, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Twenty five hundred pellets shall be packaged in a box of suitable size and design. The package shall be capable of maintaining sterility of the contents unless damaged or opened. Pellets may be packaged in a sealed bag prior to placing in the box.

5.1.2.2 Intermediate package. Thirty six unit packages shall be packaged in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed by taping the top and bottom flaps with a 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.3 Level C. Cotton pellets shall be packaged in a manner that will maintain sterility of contents and afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Two intermediate packages (72 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, Class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Two intermediate packages (72 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

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5.2.3 Level C. The packaged pellets shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

5.3 Marking.

5.3.1 Unit package. In addition to any commercial markings, each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act. The date of manufacture shall be shown in lieu of the date packed. Marking shall include the sterility lot (control) number, contract or purchase order number and the name and address of the contractor. When the manufacturer is not the contractor, the name and address of the manufacturer shall also be shown. The method of sterilization S-Steam, shall be shown following the level of packaging and date of manufacture. Example: A-MFD 1/76-S. Final sterilization shall be on or after indicated manufactured date. Marking shall also include the legend: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. Contract or purchase order number and name of the contractor shall be shown. The sterility lot (control number and method of sterilization) as specified in 5.3.1 shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. The date of manufacture shall be shown in lieu of the date packed. Marking shall include the sterility lot (control) number and the method of sterilization as specified in 5.3.1.

6. NOTES

6.1 Intended use. The cotton pellets covered by this specification are intended for use in dental procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of packaging and packing (see 5.1 and 5.2)
(c) National Stock Number (NSN)

6.3 This specification covers the following item listed in the Federal Supply Catalog:

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National Stock Number

6510-00-518-1550

Custodians:
Army - MD
Navy - MS
Air Force - 03

Item Identification

PELLET, ABSORBENT, Cotton, 2500s

Preparing activity:

Defense Supply Agency - DM

Project No. 6510-0615

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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